

1 month

ALGORITHMS FUNDAMENTALS

Introduction, Design & Types

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2026-I



UNIVERSIDAD DISTRITAL
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Outline

- 1 Introduction to Algorithms
- 2 Algorithm Design
- 3 Algorithm Types



Outline

1 Introduction to Algorithms

2 Algorithm Design

3 Algorithm Types



What is an Algorithm?

Definition

An **algorithm** is a finite sequence of well-defined **instructions** that can be mechanically executed to solve a **computational problem** or perform a task.

Key Components:

- **Input:** Zero or more quantities supplied externally
- **Output:** One or more quantities produced
- **Instructions:** Precise and unambiguous steps
- **Execution:** Can be performed mechanically

Remember

An algorithm is a *method* for solving problems, not the implementation itself!



What is an Algorithm?

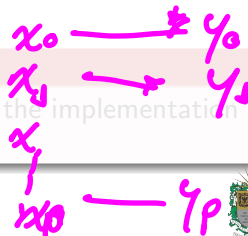
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values $\Rightarrow$ context



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design

Remember

An algorithm is a *[method for solving problems]* not the implementation itself!

x



Algorithm Characteristics

Essential Properties

- 1 **Finiteness:** Must terminate after finite number of steps
- 2 **Definiteness:** Each step must be precisely defined
- 3 **Input:** Zero or more input values
- 4 **Output:** One or more output values
- 5 **Effectiveness:** Operations must be basic and feasible

Example

Making Coffee Algorithm:

- 1 Boil water (definite action)
- 2 Add coffee grounds (precise amount)
- 3 Wait 4 minutes (finite time)
- 4 Pour into cup (effective operation)

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Making Coffee Algorithm:

- ① Boil water (definite action)
- ② Add coffee grounds (precise amount)
- ③ Wait 4 minutes (finite time)
- ④ Pour into cup (effective operation) ✓

CONSTANT P-water
temp-water
amount gr? gr/cup?

Algorithm vs. Program vs. Process

Algorithm *design*

- Abstract concept
- Language-independent
- Mathematical description
- Design phase

Program *code*

- Concrete implementation
- Language-specific
- Code representation
- Implementation phase

Process *Top Ctrl + Alt + Sppr*

- Runtime execution
- System-dependent
- Active computation
- Execution phase

Example

Example: Finding the maximum number

- **Algorithm:** "Compare each element with current maximum"
- **Program:** Python code implementing the comparison
- **Process:** Program running in memory, using CPU cycles

Algorithm vs. Program vs. Process

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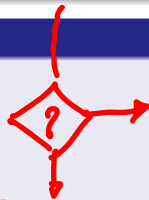
if > replace

→

Types of Algorithmic Thinking

Fundamental Approaches

- ① **Sequential Thinking:** Step-by-step execution
- ② **Conditional Thinking:** Decision-based branching
- ③ **Iterative Thinking:** Repetitive operations
- ④ **Recursive Thinking:** Self-referential solutions
- ⑤ **Parallel Thinking:** Concurrent execution

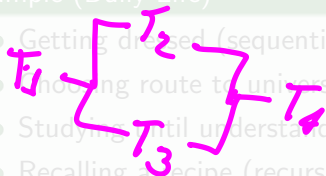


loops

automation

Example (Daily Life)

- Getting dressed (sequential)
- Choosing route to university (conditional)
- Studying until understanding (iterative)
- Recalling a recipe (recursive)
- Cooking multiple dishes (parallel)



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Study Case: Euclidean GCD Algorithm

Greatest Common Divisor Problem

Input: Two positive integers a and b

Output: The largest integer that divides both a and b

Algorithm 1 Euclidean GCD Algorithm

Input: Integers a, b where $a \geq b > 0$

while $b \neq 0$ **do**

$r \leftarrow a \bmod b$

$a \leftarrow b$

$b \leftarrow r$

end while

return a



Study Case: Euclidean GCD Algorithm

Greatest Common Divisor Problem

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Output: The largest integer that divides both a and b

Algorithm 2 Euclidean GCD Algorithm

Input: Integers a, b where $a \geq b > 0$

while $b \neq 0$ **do**

$r \leftarrow a \bmod b$

$a \leftarrow b$

$b \leftarrow r$

end while

return a

define f as $a \leftarrow$
• def f



Exercise Time! [1]

Exercise 1: Square Root Calculation

Design an algorithm to calculate the square root of a positive number using the Babylonian method:

- 1 Start with an initial guess x_0 (e.g., $S/2$ for number S)
- 2 Improve the guess: $x_{n+1} = \frac{1}{2}(x_n + \frac{S}{x_n})$
- 3 Repeat until convergence

Input: S

Exercise 2: Find Maximum in an Array

$x = S/2$

$DIFF = 0.001$

DO {

$x_{next} = (S/2) * (x + (S/x))$

$diff_x = abs(x - x_{next})$

$x = x_{next}$

} WHILE ($diff_x > DIFF$)

return x



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- 3 Repeat until convergence

if length A == 0 {
return 0.0}

Exercise 2: Find Maximum in an Array

Write an algorithm that finds the maximum element in an array of integers.

Consider: What if the array is empty? What about duplicate maximums?

Input: $[a_1, a_2, \dots, a_n]$
 $max \leftarrow A[0]$
 for $(i = 1 \rightarrow n-1)$ {
 if $(A[i] > max)$ {
 $max = A[i]$;
 }

return max



Exercise Time! [2]

Exercise 3: Algorithms in Daily Life

Identify and describe three algorithms you use in your daily life. For each:

- Define clear inputs and outputs
- List the precise steps
- Verify all algorithm characteristics



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Mathematical Notation for Algorithms [I]

Mathematical Notation

Mathematical notation provides a formal way to **express algorithms** using symbols, formulas, and structures from **mathematics**.

Why Mathematical Notation?

Universal

- **Precision:** Unambiguous specification
- **Universality:** Language-independent
- **Conciseness:** Compact representation
- **Analysis:** Enables formal verification



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Mathematical Notation for Algorithms [II]

Example

Set Operations:

- $S = \{x \in \mathbb{N} : x \text{ is prime and } x < 20\}$
- $\forall x \in S, P(x)$ (for all elements in S , property P holds)
- $\exists x \in S : Q(x)$ (there exists an element in S with property Q)

$S = \{2, 3, 5, 7, 11, 13, 17, 19\}$

Map
Filter

Example

Function Definition:

$$f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$$

$$f(a, b) = \begin{cases} a & \text{if } b = 0 \\ f(b, a \bmod b) & \text{otherwise} \end{cases}$$



Mathematical Notation for Algorithms [II]

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Set Operations:

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Euclidean GCD



Converting Mathematical Formulas to Pseudocode

Mathematical Expression

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

Another Example

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

```

1: sum ← 0
2: for i = 1 to n do
3:   sum ← sum + i2
4: end for
5: formula ←  $\frac{n(n+1)(2n+1)}{6}$ 
6: if sum = formula then
7:   return true
8: else
9:   return false
10: end if
  
```

```

1: result ← 1
2: term ← 1
3: n ← 1
4: while |term| >  $\epsilon$  do
5:   term ←  $\frac{\text{term} \cdot x}{n}$ 
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```

range (1, n+1)

```

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```

validation

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DIFF

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9: return result
  
```

Study Case: Factorial Using Different Notations

Mathematical

$$n! = \begin{cases} 1 & \text{if } n = 0 \\ n \cdot (n-1)! & \text{if } n > 0 \end{cases}$$

$$n! = \prod_{i=1}^n i$$

Iterative Pseudocode

```

1: function FACTORIAL(n)
2: result ← 1
3: for i = 1 to n do
4:   result ← result × i
5: end for
6: return result
7: end function

```

ΣΠ

↘ not > 0 ⇒ error ↗



Pseudocode Standards and Conventions

Basic Elements

• **Keywords:** if, then, else, while, for, return

• **Assignment:** $x \leftarrow \text{value}$ or $x = \text{value}$

• **Comparison:** $=, \neq, <, >, \leq, \geq$

• **Logic:** and, or, not

• **Comments:** // Single line. /* Multi-line */

reservadas

type(var)
type-

C1	C2	C3	C4
V	V	V	V
V	F	V	V
F	F	F	F

bool $\rightarrow$ $\neq$ $==$
 $<$ $<=$ $>$ $>=$

comp and comp
" or " ?

not comp

Algorithm Documentation Best Practices

Essential Documentation Elements

- ① **Purpose:** What does the algorithm do? *→ this function...*
- ② **Preconditions:** What must be true before execution? *for $n \geq 0$
even $a > b$*
- ③ **Postconditions:** What is guaranteed after execution?
- * ④ **Parameters:** Input and output specifications *Args: Returns:*
- ⑤ **Complexity:** Time and space requirements
- ⑥ **Examples:** Sample inputs and outputs

Doc.

0

1	-
2	-
3	-

Programming
Caribest



Study Case: Fibonacci Sequence

Problem Definition

The Fibonacci sequence: $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$

Purpose: Compute the n th Fibonacci number

Preconditions: $n \geq 0$ (non-negative integer)

Postconditions: Returns F_n where $F_0 = 0$, $F_1 = 1$

Parameters:

- n : position in Fibonacci sequence

Complexity: $O(n)$ time, $O(1)$ space

Example: FIBONACCI ITERATIVE(5) = 5

$$f(n) = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ f(n-1) + f(n-2) & n \geq 2 \end{cases}$$

$$\text{for } i \text{ from } 2 \text{ to } n \\ f(i) = f[i-1] + f[i-2]$$



Study Case: Fibonacci Sequence

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Postconditions: Returns F_n where $F_0 = 0, F_1 = 1$

Parameters:

- n : position in Fibonacci sequence $\rightarrow$ 7-th

Complexity: $O(n)$ time, $O(1)$ space

Example: FIBONACCIITERATIVE(5) = 5

0 0
1 1
1 2
2 3
3 5
5 8



Control Structures

Sequential

- 1: Step 1
- 2: Step 2
- 3: Step 3
- 4: ...

Execution: One after another

Typical

Conditional

- 1: **if** condition **then**
- 2: Action A
- 3: **else**
- 4: Action B
- 5: **end if**

Execution: Based on condition

Iterative

- 1: **while** condition **do**
- 2: Action
- 3: **end while**
- 4: **for** i = 1 to n **do**
- 5: Action
- 6: **end for**

Execution: Repeated



Control Structures

Sequential

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- 4: end while

- 2: **for** i = 1 to n **do**
- 3: Action
- 4: end for

Execution: Repeated

< > ≤ ≥ ≠ ==

Java - C

$x = y > 5 ? 3 : 4;$

Python

$x = 3$ if $y > 5$ else 4

if ($y > 5$) {
 $x = 3;$
} else {
 $x = 4;$
}

$2 == 2.0$

↳ true

↳ false



Control Structures

Sequential

- 1: Step 1
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Execution: One after another

Conditional

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Iterative

- 1: **while** condition **do**
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- 4: **for** $i = 1$ to n **do**
- 5: Action
- 6: **end for**

Execution: Repeated



Control Flow Concepts

- **Entry Point:** Where execution begins
- **Exit Point:** Where execution ends
- **Decision Points:** Where flow branches
- **Loop Control:** How iterations are managed

inputs - pre conditions
~ output

break

continue



Study Case: Decision-Making Algorithms

Grade Classification Algorithm

Problem: Convert numerical grade to letter grade

Decision Tree

- *Score* ≥ 90 ? $\rightarrow A$
- *Score* ≥ 80 ? $\rightarrow B$
- *Score* ≥ 70 ? $\rightarrow C$
- *Score* ≥ 60 ? $\rightarrow D$
- *Otherwise* $\rightarrow F$

Large

Nom $\rightarrow$ Letter

- Mutually exclusive conditions
- Order matters in if-elseif chains
- Default case handling



Study Case: Decision-Making Algorithms

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Problem: Convert numerical grade to letter grade

Decision Tree

- else
 else
 else
- Score $\geq 90?$ $\rightarrow$ A
 - Score $\geq 80?$ $\rightarrow$ B
 - Score $\geq 70?$ $\rightarrow$ C
 - Score $\geq 60?$ $\rightarrow$ D
 - Otherwise $\rightarrow$ F

switch

Key Concepts

- **Mutually exclusive** conditions
- **Order matters** in if-elsif chains
- **Default case** handling

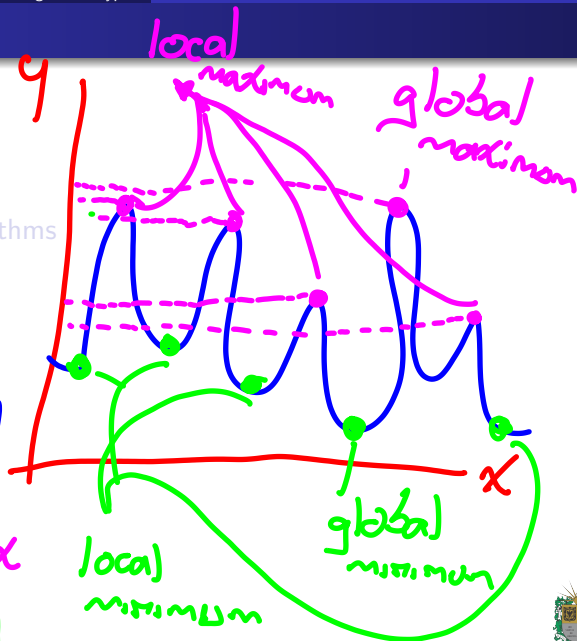


Outline

1 Introduction to Algorithms

2 Algorithm Design

3 **Algorithm Types**



$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 x^0$$



Greedy Algorithms: Philosophy and Approach

Definition

A greedy algorithm makes locally optimal choices at each step, hoping to find a global optimum.

Greedy Strategy

- 1 **Greedy Choice:** At each step, choose the **best available** option
- 2 **Optimal Substructure:** Optimal solution contains optimal solutions to subproblems
- 3 **No Backtracking:** Never reconsider previous choices



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J. most chapters Greedy not ch...

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Greedy Algorithms: When to Use Them

When Greedy Works:

- Activity selection
- Minimum spanning trees
- Huffman coding

→ wave

Compression

text 18 bytes

~ ~ PATTERN

^ ^ - ^ ^ - ^ ^ - ^ ^ -

-- PATTERN

~ ~ ~ ~ ~
~ ~ ~ ~ ~
~ ~ ~ ~ ~

1. all words

2. sort words by size

2.1 sort by occurrences

PATTERN →



Greedy Algorithms: When to Use Them

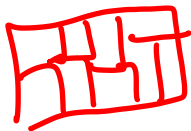
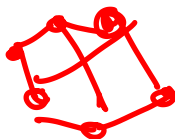
When Greedy **Works**:

- Activity selection
- Minimum spanning trees
- Huffman coding

When Greedy **Fails**:

- 0/1 Knapsack problem
- Traveling salesman
- Graph coloring

✓ 5 → ele v p



Study Case: Coin Change Problem

Problem Statement

Given: Coin denominations $[d_1, d_2, \dots, d_k]$ and amount n

Goal: Find minimum number of coins to make amount n

1000 - 500 - 200 - 100 - 50

1. sort denominations $\rightarrow$ great to less

2. loop $\rightarrow$ coins = $n / d_i \rightarrow$ integer

$n = n - (\text{coins} * d_i)$



Study Case: Coin Change Problem [Solution]

Denominations: [25, 10, 5, 1] cents, **Amount:** 67 cents

```
1: Algorithm GREEDYCOINCHANGE(denominations, amount)
2: result  $\leftarrow []$ 
3: for each denomination d in descending order do
4:   while amount  $\geq d$  do
5:     result.append(d)
6:     amount  $\leftarrow$  amount  $- d$ 
7:   end while
8: end for
9: return result
```

Solution: $67 = 25 + 25 + 10 + 5 + 1 + 1$ (6 coins)



Divide and Conquer: Methodology

Definition

Divide and Conquer breaks a problem into smaller subproblems, solves them recursively, then combines solutions.

Handwritten notes: A blue circle around "Divide and Conquer", a red circle around "a problem", and a red arrow pointing from "a problem" to "smaller subproblems". The word "similar" is written in red cursive.

Three-Step Process

- 1 **Divide:** Break problem into smaller subproblems
- 2 **Conquer:** Solve subproblems recursively
- 3 **Combine:** Merge solutions to get final answer



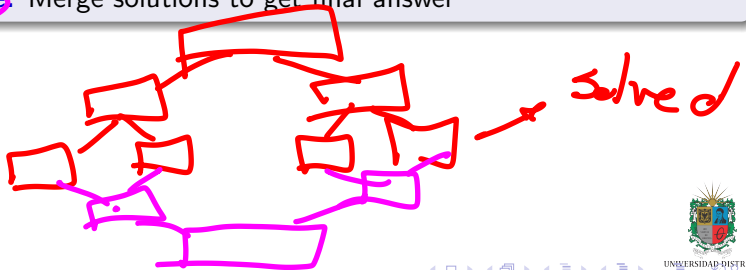
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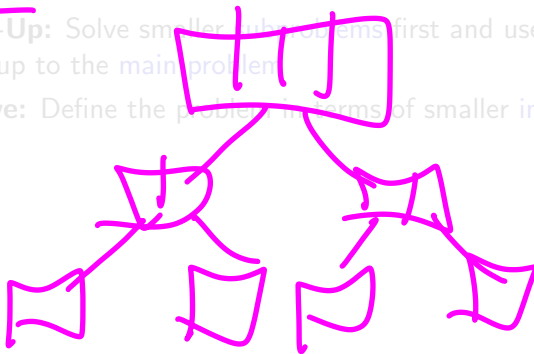
Divide and Conquer: Algorithm Template

```
1: Algorithm DIVIDEANDCONQUER(problem)
2: if problem is small enough then
3:   return SOLVEDIRECTLY(problem)
4: else
5:   subproblems  $\leftarrow$  DIVIDE(problem)
6:   for each subproblem do
7:     solution  $\leftarrow$  DIVIDEANDCONQUER(subproblem)
8:   end for
9:   return COMBINE(solutions)
10: end if
```



Problem Decomposition Strategies

- **Top-Down:** Start with the main problem and break it down into subproblems
- **Bottom-Up:** Solve smaller subproblems first and use their solutions to build up to the main problem
- **Recursive:** Define the problem in terms of smaller instances of itself



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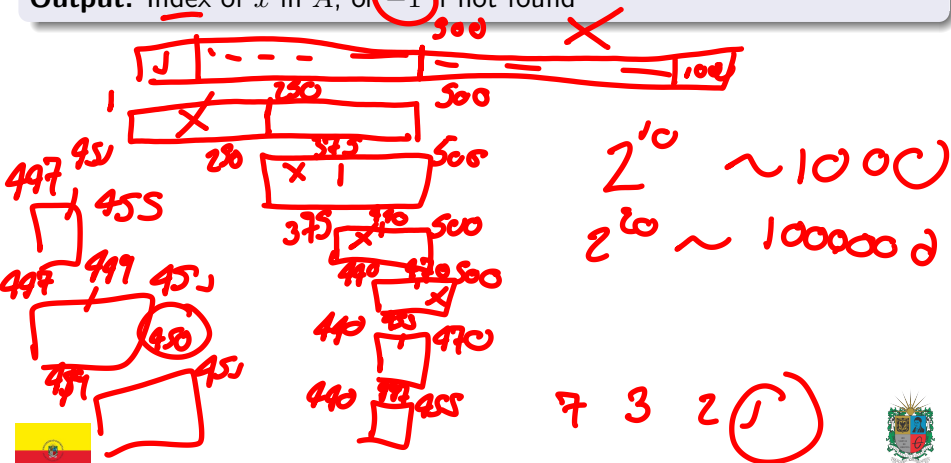


Study Case: Binary Search

Problem

Input: Sorted array $A[1..n]$ and search value x

Output: Index of x in A , or -1 if not found

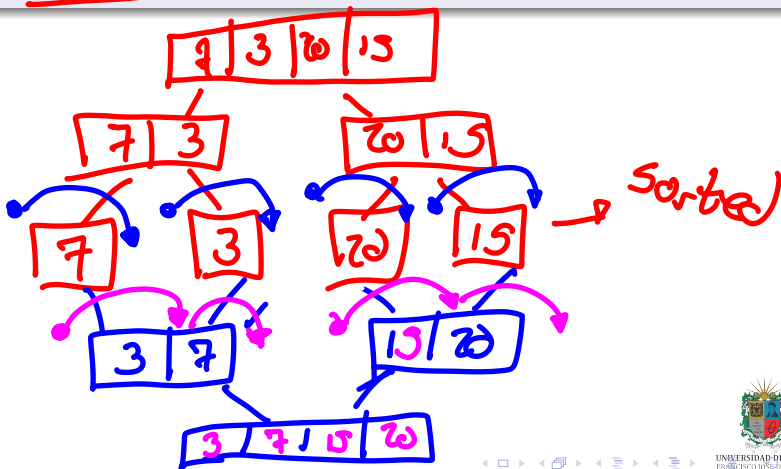


Study Case: Merge Sort Approach

Sorting Problem

Input: Array $A[1..n]$ of comparable elements

Output: Array sorted in ascending order



Demo Time! Clean the House Exercise

Problem

You need to clean your house efficiently. You have different rooms and different cleaning tasks.

Greedy Strategy: Always clean the dirtiest room first

- 1 Assess dirt level of all rooms
- 2 Choose room with highest dirt level
- 3 Clean that room completely
- 4 Repeat until all rooms clean

Advantage: Maximum immediate impact

Divide & Conquer Strategy:
Split house systematically

- 1 Divide house into sections (floors/wings)
- 2 Recursively clean each section
- 3 Within each room, divide into areas
- 4 Clean areas systematically

Advantage: Systematic coverage



Demo Time! Clean the House Exercise

Problem

You need to clean your house efficiently. You have different rooms and different cleaning tasks.

Greedy Strategy: Always clean the dirtiest room first

- ① Assess dirt level of all rooms
- ② Choose room with highest dirt level
- ③ Clean that room completely
- ④ Repeat until all rooms clean

Advantage: Maximum immediate impact



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Exercise: Knapsack Problem

Problem Statement

Given: Knapsack capacity W , items with weights w_i and values v_i

Goal: Maximize total value without exceeding weight capacity



Exercise: Knapsack Problem [Solution]

0/1 Knapsack (Greedy Approach)

Cannot take fractions - either take entire item or leave it

- **Strategy:** Sort by value/weight ratio, take whole items
- **Optimal:** No! (Greedy doesn't guarantee optimal solution)

- 1: Sort items by v_i/w_i descending
- 2: **for** each item i **do**
- 3: **if** $w_i \leq$ remaining capacity **then**
- 4: Take entire item
- 5: Update remaining capacity
- 6: **end if**
- 7: **end for**



Brute Force: Exhaustive Search

Definition

Brute force algorithms try **all possible solutions** until finding the correct one.

Characteristics:

- **Exhaustive:** Examines every possibility
- **Guaranteed:** Always finds optimal solution (if exists)
- **Expensive:** Often exponential time complexity
- **Simple:** Easy to understand and implement



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Brute Force: When to Use It

- 1 Problem **size** is **small**
- 2 **No efficient** algorithm is known
- 3 **Correctness** is more important than efficiency
- 4 As **baseline** for comparing other algorithms
- 5 When **optimization** overhead exceeds brute force cost



Study Case: Password Cracking

Problem

Given: Encrypted password hash and character set

Goal: Find the original password



Study Case: Password Cracking [Solution]

```
1: Algorithm BRUTEFORCEPASSWORD(hash, maxlength)
2: for length = 1 to maxlength do
3:   for each possible string s of given length do
4:     if HASH(s) = hash then
5:       return s // Password found!
6:     end if
7:   end for
8: end for
9: return null // Password not found
```

Character set: [a-z] (26 characters)

Password length: 8 characters

Total attempts: $26^8 \approx 208$ billion

Time estimate: Weeks on single computer!



Study Case: Password Cracking [Solution]

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Backtracking: Systematic Trial and Error

Definition

Backtracking explores solution space systematically, **abandoning partial solutions** that cannot lead to valid solutions.

Backtracking Strategy

- 1 **Choose:** Make a choice from available options
- 2 **Explore:** Recursively explore consequences
- 3 **Unchoose:** If path fails, backtrack and try another



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Backtracking: Algorithm

```

1: Algorithm BACKTRACK(partial_solution)
2: if ISCOMPLETE(partial_solution) then
3:   return partial_solution
4: end if
5: for each possible choice c do
6:   if ISVALID(partial_solution, c) then
7:     partial_solution.add(c)
8:     result  $\leftarrow$  BACKTRACK(partial_solution)
9:     if result  $\neq$  null then
10:      return result
11:   end if
12:   partial_solution.remove(c) // Backtrack!
13: end if
14: end for
15: return null
  
```



Study Case: N-Queens Problem

Problem

Place N queens on an $N \times N$ chessboard such that no two queens attack each other.



Study Case: N-Queens Problem [Solution]

```

1: Algorithm SOLVENQUEENS(board, row)
2: if row = N then
3:   return board // Solution found!
4: end if
5: for each column col in 0 to N - 1 do
6:   if ISSAFE(board, row, col) then
7:     board[row][col]  $\leftarrow$  Q // Place queen
8:     result  $\leftarrow$  SOLVENQUEENS(board, row + 1)
9:     if result  $\neq$  null then
10:      return result
11:    end if
12:    board[row][col]  $\leftarrow$  . // Remove queen (backtrack)
13:  end if
14: end for
15: return null // No solution found

```



Implementation Patterns for Backtracking

- **Recursive Structure:** Backtracking algorithms are typically implemented using recursion.
- **State Representation:** Maintain a representation of the current state (e.g., partial solution).
- **Choice Points:** Identify points where decisions are made (choices to explore).
- **Validity Checks:** Implement checks to determine if the current state is valid.
- **Base Case:** Define a base case for when a complete solution is found.
- **Backtracking Mechanism:** Ensure that the algorithm can revert to previous states when necessary.



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Summary: Algorithm Types

Key Takeaways

- **Greedy:** Make locally optimal choices (works when greedy choice property holds)
- **Divide & Conquer:** Break into subproblems, solve recursively, combine solutions
- **Brute Force:** Try all possibilities (guarantees correctness but expensive)
- **Backtracking:** Systematic search with pruning (intelligent brute force)



Summary: Algorithm Types

Algorithm Selection Guide

- **Small problem size?** → Consider brute force
- **Optimization problem with greedy choice property?** → Try greedy
- **Problem naturally divides into subproblems?** → Use divide & conquer
- **Constraint satisfaction or search problem?** → Apply backtracking



Outline

- 1 Introduction to Algorithms
- 2 Algorithm Design
- 3 Algorithm Types



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-i>

